

$$r = 20 \, \Omega$$

23 Ans: A

The voltmeter is connected in parallel to the combined resistors (P and Q).
The voltmeter can be considered to be connected directly to the battery as its resistance is very large compared to P and Q ($10 \, \Omega$ each). Hence, the p.d. as measured by the voltmeter will always register the value of 4 V irrespective of the position of the slider.

24 Ans: D